# Kernix 0.4.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

**13** workloads

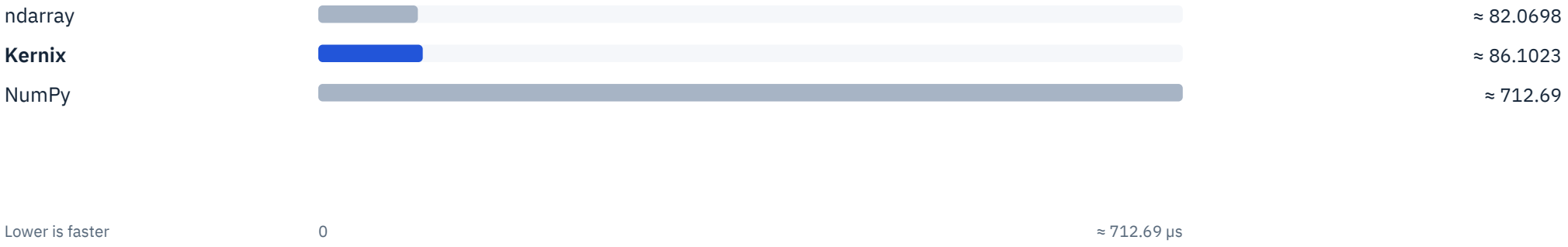
**38** observations

Source precision retained

## Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




### How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

# Kernix 0.4.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product

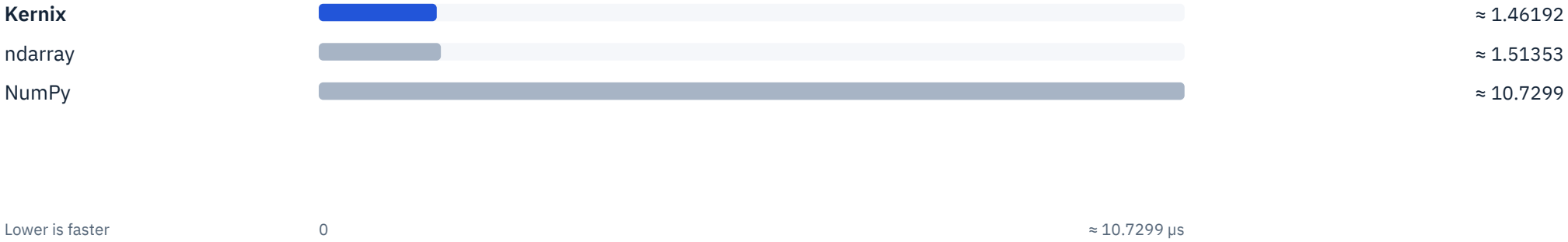
# Kernix 0.4.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Matrix multiplication · reused output

Batch-mean median

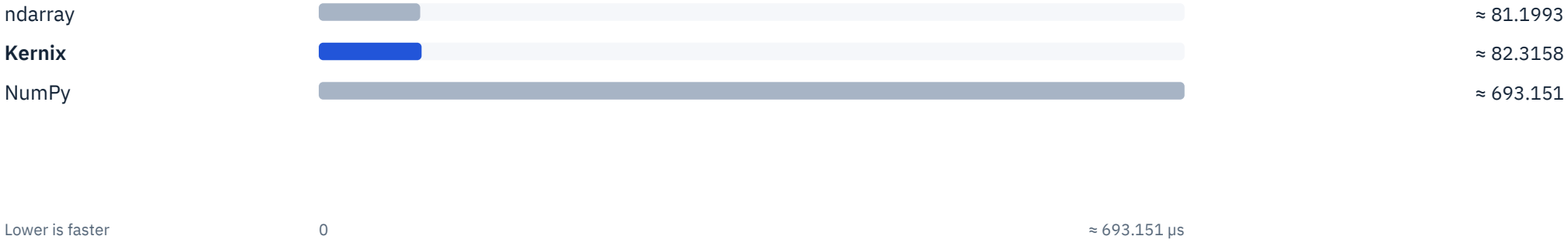
32 × 32 · 32 × 32 · f64 · 1 worker · reused output




## Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output



# Kernix 0.4.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Cholesky factorization

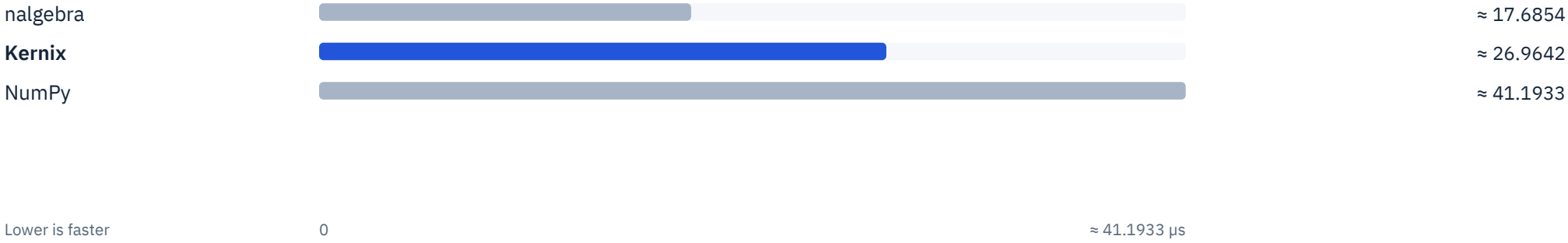
Batch-mean median

64 × 64 · f64 · 1 worker · allocated output

## Linear system solve

Batch-mean median

64 × 64 · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



# Kernix 0.4.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Thin QR

Batch-mean median

96 × 32 · f64 · 1 worker · allocated output

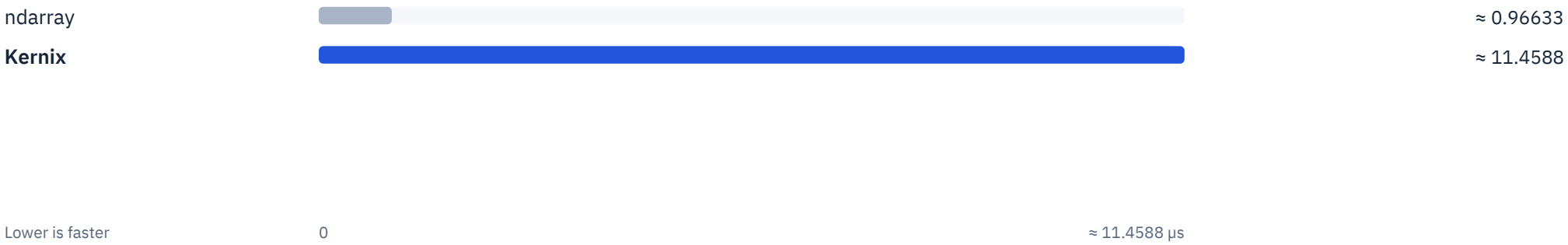
# Kernix 0.4.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Dense training step

Batch-mean median

32 × 16 × 8 · f64 · Graph-based Kernix vs hand-derived comparator updates; API costs differ



# Kernix 0.4.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Elementwise addition

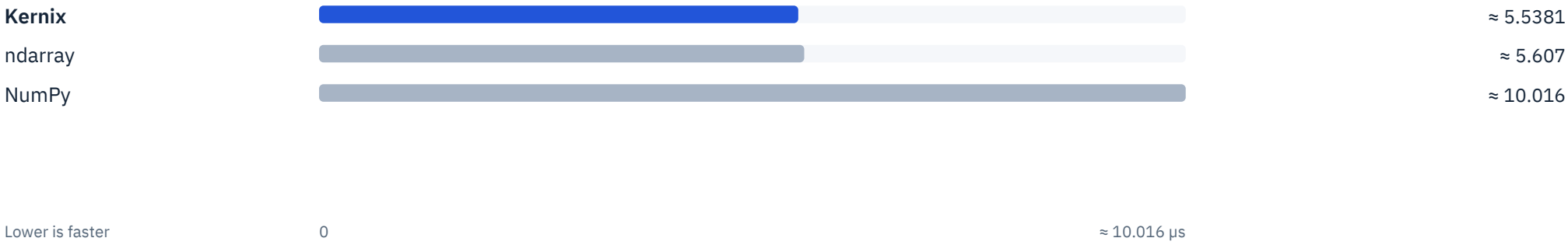
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

## Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



# Kernix 0.4.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median

# Kernix 0.4.0 / Vision workloads

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

## RGB resize & normalization

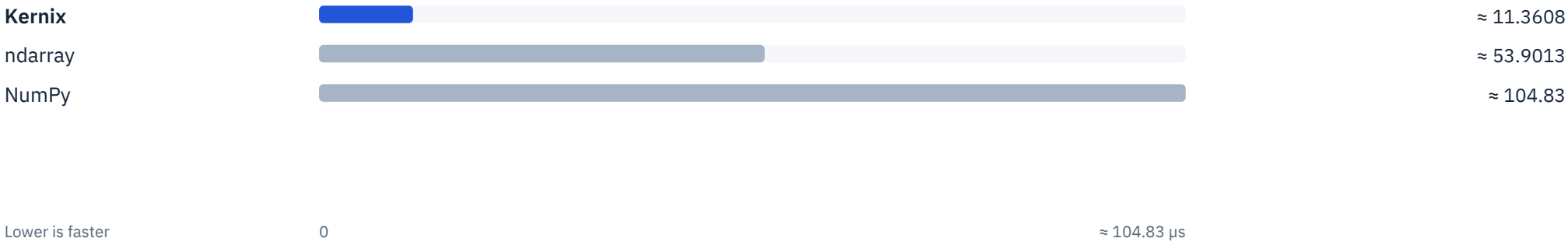
Batch-mean median

1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 · 1 worker · allocated output · comparator implementations differ

## 2D convolution

Batch-mean median

1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 · 1 worker · allocated output · comparator implementations differ



# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1600.2428470844268 | 1.6002428470844268 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 1416.7458491528305 | 1.4167458491528305 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 10639.569787234042 | 10.639569787234042 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 86102.32989690722 | 86.10232989690722 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 82069.83934426229 | 82.06983934426229 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 712689.7777777778 | 712.6897777777778 |
| **Matrix multiplication · reused output / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1461.9173147769136 | 1.4619173147769136 |
| **Matrix multiplication · reused output / ndarray**<br>32 × 32 · 32 × 32 · f64 | 1513.5317229688824 | 1.5135317229688824 |
| **Matrix multiplication · reused output / NumPy**<br>32 × 32 · 32 × 32 · f64 | 10729.9141999142 | 10.7299141999142 |
| **Matrix multiplication · reused output / Kernix**<br>128 × 128 · 128 × 128 · f64 | 82315.77302631579 | 82.31577302631579 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication · reused output / ndarray**<br>128 × 128 · 128 × 128 · f64 | 81199.30844155845 | 81.19930844155845 |
| **Matrix multiplication · reused output / NumPy**<br>128 × 128 · 128 × 128 · f64 | 693150.5675675676 | 693.1505675675676 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 23374.6476635514 | 23.3746476635514 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 10564.963244613435 | 10.564963244613435 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 26421.718057022175 | 26.421718057022175 |
| **Linear system solve / Kernix**<br>64 × 64 · f64 | 26964.24676724138 | 26.96424676724138 |
| **Linear system solve / nalgebra**<br>64 × 64 · f64 | 17685.366336633662 | 17.685366336633662 |
| **Linear system solve / NumPy**<br>64 × 64 · f64 | 41193.26194398682 | 41.19326194398682 |
| **Thin QR / Kernix**<br>96 × 32 · f64 | 88613.39929328622 | 88.61339929328622 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 32263.832258064514 | 32.263832258064514 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / NumPy**<br>96 × 32 · f64 | 79375.3746031746 | 79.3753746031746 |
| **Dense training step / Kernix**<br>32 × 16 × 8 · f64 | 11458.840971585701 | 11.458840971585701 |
| **Dense training step / ndarray**<br>32 × 16 × 8 · f64 | 966.3295068027211 | 0.9663295068027211 |
| **Elementwise addition / Kernix**<br>65,536 elements · f64 | 21233.55432937182 | 21.23355432937182 |
| **Elementwise addition / ndarray**<br>65,536 elements · f64 | 22836.65296803653 | 22.83665296803653 |
| **Elementwise addition / NumPy**<br>65,536 elements · f64 | 22884.931381518756 | 22.884931381518756 |
| **Sum reduction / Kernix**<br>65,536 elements · f64 | 5538.095459579181 | 5.538095459579181 |
| **Sum reduction / ndarray**<br>65,536 elements · f64 | 5607.004709576138 | 5.607004709576138 |
| **Sum reduction / NumPy**<br>65,536 elements · f64 | 10016.018822587104 | 10.016018822587104 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 77015.27076923077 | 77.01527076923077 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Row softmax / ndarray**<br>256 × 64 · f64 | 69555.83888888889 | 69.55583888888889 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 94432.09056603773 | 94.43209056603773 |
| **RGB resize & normalization / Kernix**<br>1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 | 438953.1379310345 | 438.9531379310345 |
| **RGB resize & normalization / ndarray**<br>1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 | 685549.3513513514 | 685.5493513513514 |
| **RGB resize & normalization / NumPy**<br>1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 | 3994365.4285714286 | 3994.3654285714286 |
| **2D convolution / Kernix**<br>1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 | 11360.770104497955 | 11.360770104497955 |
| **2D convolution / ndarray**<br>1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 | 53901.27370689655 | 53.90127370689655 |
| **2D convolution / NumPy**<br>1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 | 104830.2510460251 | 104.8302510460251 |